Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

Presentation Title

Floatsat Winter Semester 2025-26 Group II

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Example Slide

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Introduction

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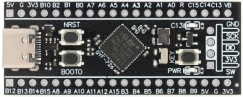
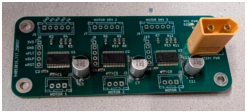
Electrical Design

Main electrical component used to design this project are.

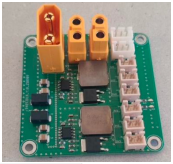
- ◇ Two microcontrollers
  - ✓ STM32F401: Primary Controller with extensive GPIO.
  - ✓ DFRobot Beetle ESP32 C6: For wireless communication with ground station.
- ◇ Motor Driver : Receives PWM control signals from the STM32 and uses the high-power battery supply to generate controlled three-phase outputs for regulating the motor speed.
- ◇ 3x T-Motor GB2208:Three phase motors act as reaction wheels, producing controlled torques to regulate the FloatSat's roll, pitch, and yaw orientation.





Electrical Design

- Sparkfun LSM9DS1: 9-DOF IMU that measures the motion and orientation of the FloatSat by combining accelerometer, gyroscope, and magnetometer data.
- Battery Management Board: Power management module supplying regulated 3.3 V, 5 V, and 12 V outputs to different subsystems.
- Lithium-Polymer Battery



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Electrical Design

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Figure: FloatSat system

Software Design

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Thank You for Your Attention.

Are there any Questions?

References

[HZZ+20] Stefan Herrnleben, Bernd Zeidler, Marwin Züfle, Christian Krupitzer, and Samuel Kounev.
A Concept for Crowd-sensed Prediction of Mobile Network Connectivity.
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